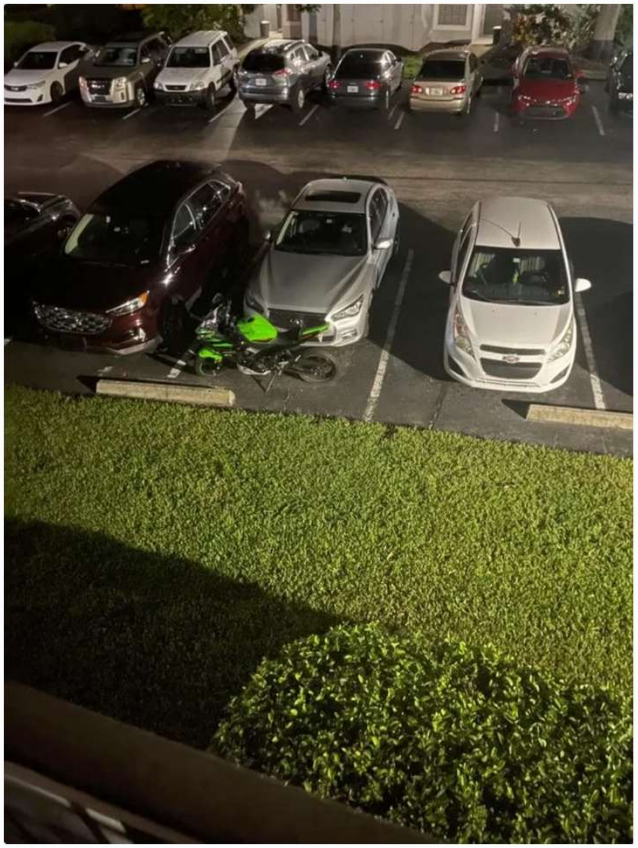
PROJECT TITLE



Parking Lot Management System

Using Core Java & Object-Oriented Programming Concepts

Project Details

Subject: Object Oriented Programming with Java
Project ID: RU-100-01-00012

Student Information

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College: Rungta International Skills University
Department: Computer Science & Engineering
Academics Year: 2025-29 (First Year)

INTRODUCTION SMART MOBILITY

Introduction

Addressing the growing demand for smart and efficient urban mobility solutions.

Problems in Traditional Systems

- Manual parking slot management
- Time-consuming vehicle tracking
- Difficulty in calculating parking fees
- Lack of real-time slot availability
- Human errors and inefficiency
- Poor customer experience

Why Automation Matters

- Efficient slot allocation
- Automated fee calculation
- Real-time availability tracking
- Reduced manual workload
- Improved accuracy
- Better customer satisfaction

OBJECTIVES | SCOPE | SMART PARKING

Objectives & Scope

Key goals and application areas for the smart parking system.

Objectives

- Manage parking slots efficiently
- Assign parking spaces automatically
- Track vehicle entry and exit
- Calculate parking fees accurately
- Reduce manual work and human errors
- Apply OOP concepts in Java implementation

Scope

- Suitable for malls and shopping centers
- Applicable to office buildings and corporate parks
- Ideal for colleges and educational institutions
- Effective for small to medium parking areas
- Can be expanded for larger applications
- Scalable architecture for future enhancements

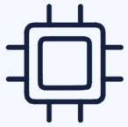
SYSTEM REQUIREMENTS | TECHNOLOGIES | SMART PARKING

System Requirements & Technologies

Essential hardware, software, and core technologies.

Hardware Requirements

4GB RAM minimum



Intel i5 or equivalent

500MB free storage

1024×768 display or higher

Software Requirements

1. **Java JDK 8 or higher**
   Required for compiling and running the application.
2. **Code Editor / IDE**
   VS Code, Eclipse, or IntelliJ IDEA for development.
3. **Operating System**
   Compatible with Windows, Linux, or macOS environments.
4. **Git Version Control**
   Used for source control and project collaboration.

Technologies Used

Core Java
Object-oriented foundation for the full system.

OOP Concepts
Encapsulation, inheritance, and polymorphism.

Collections
ArrayList and the Collections Framework for data handling.

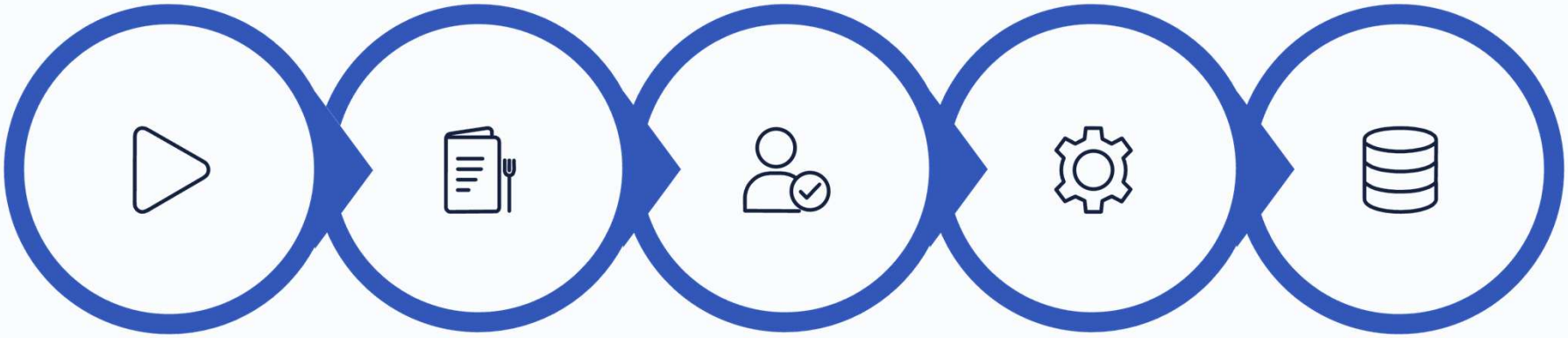
File I/O
Persistent storage and retrieval of parking data.

Exception Handling
Improves stability and error management.

WORKFLOW | PROCESS FLOW | SMART PARKING

System Workflow & Process Flow

Simple, sequential steps from start to finish.



Start | Display Menu | User Chooses | Process Tasks | Store Info

The workflow moves step by step through a continuous cycle, returning users to the main menu whenever they choose to continue.

CLASS DIAGRAM | OOP CONCEPTS | MINIMAL DESIGN

Class Diagram & OOP Concepts

Minimalist class structure and core OOP principles.

Class Structure (UML-style)

Vehicle Class:

```
- vehicleId: String
- vehicleName: String
- entryTime: LocalDateTime
- exitTime: LocalDateTime
+ getVehicleDetails()
+ calculateDuration()
```

ParkingLot Class:

```
- totalSlots: int
- availableSlots: int
- vehicles: ArrayList<Vehicle>
+ parkVehicle()
+ removeVehicle()
+ searchVehicle()
+ getAvailableSlots()
```

Main Class:

```
- displayMenu()
- handleUserInput()
- main()
```

OOP Concepts Explained

Encapsulation: Data hiding with private variables and public methods

Abstraction: Hiding complex implementation details

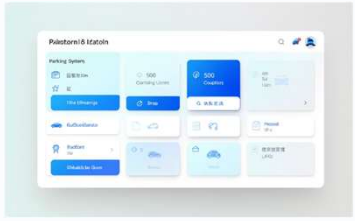
Inheritance: Extending classes for code reuse

Polymorphism: Method overriding and overloading

PROJECT DEMO | PARKING SYSTEM | UI WALKTHROUGH

A modern walkthrough of the parking system interface.

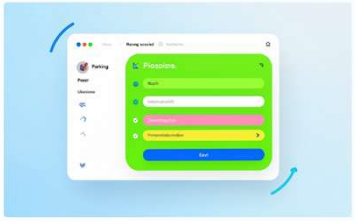
Follow the user journey across the main screens to see how parking, searching, status checks, and vehicle removal work together in a smooth, intuitive flow.



BLUE

Main Menu Screen

The central starting point for the system, giving users quick access to every major parking action in one clean, organized view.



GREEN

Vehicle Parking Screen

Capture vehicle details, assign an available slot, and complete the parking process efficiently with a clear, guided form.

ORANGE

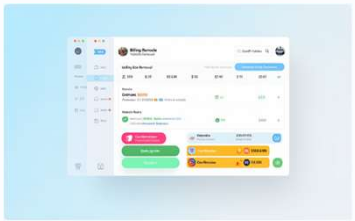
Vehicle Search Screen

Locate parked vehicles instantly using an ID or plate number, making retrieval and verification fast, accurate, and hassle-free.

PURPLE

Parking Status Screen

Monitor current occupancy, see parked vehicles at a glance, and review how many spaces remain available in real time.



RED

Vehicle Removal Screen

Remove vehicles smoothly while automatically calculating the final charge, helping the checkout experience feel simple and professional.

CONCLUSION | FUTURE SCOPE | PROJECT SUMMARY

Conclusion & Future Scope

A polished summary of today's results and the next wave of improvements.

This final section highlights the parking system's current strengths, then previews the features that can make it even smarter, faster, and more connected in the future.

Key Achievements

Streamlined parking operations

Organized the core workflow for parking, searching, status checks, and vehicle removal.

Reduced manual effort

Automated repetitive steps to save time and minimize human error.

Improved slot tracking

Made space availability easier to monitor and manage in real time.

Applied Java OOP concepts

Used an object-oriented structure that supports clean code and maintainability.

Created a user-friendly interface

Designed simple screens that make common operations quick and intuitive.

Built a scalable foundation

Prepared the project for future features, integrations, and growth.

Future Enhancements

Online booking system

Enable advance reservations for better convenience and planning.

QR code entry

Speed up access with contactless verification at entry and exit.

Payment gateway integration

Support secure digital payments and faster checkout.

Mobile application

Extend system access to phones for anytime, anywhere use.

Real-time monitoring

Provide live visibility into parking availability and status.

Analytics and notifications

Add reports, insights, and alerts to improve decision-making.

Thank You